

Lecture 22: Practice Problems — Comprehensive Solutions Guide

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Executive Architecture & Engineering Standards

This document provides production-grade reference solutions, formal mathematical proofs, and executive visual engineering implementations for the 3 industry case studies in **Lecture 22**.

Solution to Case 1: Gradient Descent Optimization on Non-Convex & Quadratic Loss Landscapes

1. Mathematical Derivations & Proofs

Task 1: Gradient & Recurrence Relation

For quadratic loss $\mathcal{L}(w) = 2(w - 3)^2 + 4$:

$$\frac{d\mathcal{L}}{dw} = 4(w - 3)$$

The Gradient Descent recurrence relation is:

$$w_{t+1} = w_t - \eta \frac{d\mathcal{L}}{dw}(w_t) = w_t - 4\eta(w_t - 3)$$

Subtract 3 from both sides to analyze distance from optimal $w^* = 3$:

$$w_{t+1} - 3 = (1 - 4\eta)(w_t - 3)$$

By induction, the error after t steps is:

$$e_t = (w_t - 3) = (1 - 4\eta)^t(w_0 - 3)$$

Task 2: Stability & Convergence Analysis

For the error to converge to zero as $t \rightarrow \infty$, the contraction factor must satisfy $|1 - 4\eta| < 1$:

$$-1 < 1 - 4\eta < 1 \implies -2 < -4\eta < 0 \implies \boxed{0 < \eta < 0.5}$$

Behavior breakdown:

1. **Monotonic Convergence** ($0 < \eta < 0.25$): $0 < 1 - 4\eta < 1$. No oscillations.
2. **Deadbeat Optimal Convergence** ($\eta = 0.25$): $1 - 4(0.25) = 0 \implies w_1 = 3$ in a single step!
3. **Oscillatory Convergence** ($0.25 < \eta < 0.5$): $-1 < 1 - 4\eta < 0$. Error alternates signs but decays.
4. **Divergent Explosion** ($\eta > 0.5$): $|1 - 4\eta| > 1$. Error exponentially amplifies to $\pm\infty$.

Task 3: Non-Convex Loss Extrema Analysis

For $\mathcal{J}(w) = w^4 - 4w^2 + 2w$:

$$\mathcal{J}'(w) = 4w^3 - 8w + 2 = 2(2w^3 - 4w + 1) = 0$$

The roots are approximately:

- $w_1 \approx -1.525$ (Global Minimum: $\mathcal{J} \approx -6.91$)
- $w_2 \approx 0.260$ (Local Maximum: $\mathcal{J} \approx 0.25$)
- $w_3 \approx 1.265$ (Local Minimum: $\mathcal{J} \approx -1.34$)

Evaluating the second derivative $\mathcal{J}''(w) = 12w^2 - 8$:

- $\mathcal{J}''(-1.525) \approx 12(2.326) - 8 = 19.91 > 0 \implies$ Local Minimum.
- $\mathcal{J}''(0.260) \approx 12(0.068) - 8 = -7.18 < 0 \implies$ Local Maximum.
- $\mathcal{J}''(1.265) \approx 12(1.600) - 8 = 11.20 > 0 \implies$ Local Minimum.

Initialization at $w_0 = -2$ leads to global minimum $w \approx -1.525$, whereas $w_0 = +2$ is trapped in sub-optimal local minimum $w \approx 1.265$.

Solution to Case 2: Deep Learning Activation Derivatives & the Vanishing Gradient Dilemma

1. Mathematical Derivations & Proofs

Task 1: Sigmoid Derivative Proof & Upper Bound

For $\sigma(z) = (1 + e^{-z})^{-1}$:

$$\sigma'(z) = -(1 + e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

Rewrite:

$$\sigma'(z) = \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}} = \sigma(z) \cdot \left(\frac{1 + e^{-z} - 1}{1 + e^{-z}} \right) = \boxed{\sigma(z)(1 - \sigma(z))}$$

To find the maximum of $g(s) = s(1 - s)$ where $s = \sigma(z) \in (0, 1)$:

$$g'(s) = 1 - 2s = 0 \implies s = \frac{1}{2} \implies \sigma(z) = \frac{1}{2} \iff z = 0$$

The maximum derivative is:

$$\max_z \sigma'(z) = \frac{1}{2} \left(1 - \frac{1}{2} \right) = \boxed{\frac{1}{4} = 0.25}$$

Task 2: Tanh Derivative Proof

For $\tanh(z) = \frac{\sinh(z)}{\cosh(z)}$:

$$\tanh'(z) = \frac{\cosh(z) \cosh(z) - \sinh(z) \sinh(z)}{\cosh^2(z)} = \frac{\cosh^2(z) - \sinh^2(z)}{\cosh^2(z)} = \boxed{1 - \tanh^2(z)}$$

Since $\tanh^2(z) \geq 0$, the maximum occurs when $\tanh(z) = 0 \implies z = 0$:

$$\max_z \tanh'(z) = 1 - 0^2 = \boxed{1.0}$$

Task 3: Gradient Attenuation Across 10 Layers

In an $L = 10$ layer network with Sigmoid activations:

$$\prod_{l=1}^{10} \sigma'(z^{(l)}) \leq \prod_{l=1}^{10} (0.25) = (0.25)^{10} = \frac{1}{4^{10}} = \frac{1}{1,048,576} \approx 9.537 \times 10^{-7}$$

Gradients reaching the input layer are attenuated by over **one million times**, rendering weight updates completely stalled.

Task 4: Why ReLU Eliminates Gradient Attenuation

For Rectified Linear Unit:

$$\text{ReLU}(z) = \max(0, z) \implies \text{ReLU}'(z) = \begin{cases} 1.0 & z > 0 \\ 0.0 & z < 0 \end{cases}$$

For all active neurons ($z > 0$), $\text{ReLU}'(z) = 1.0$. The gradient product becomes:

$$\prod_{l=1}^{10} 1.0 = 1.0$$

There is zero multiplicative decay, allowing deep networks to scale to hundreds of layers.

Solution to Case 3: Higher-Order Calculus & Curvature: Complete Polynomial Extremum Analysis

1. Mathematical Derivations & Proofs

Task 1: First Derivative & Critical Points

For $f(x) = x^3 - 6x^2 + 9x$:

$$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$$

Setting $f'(x) = 0$:

$$(x - 1)(x - 3) = 0 \implies x_1 = 1, \quad x_2 = 3$$

Task 2: Second Derivative Classification

Differentiating again:

$$f''(x) = 6x - 12 = 6(x - 2)$$

1. At $x_1 = 1$:

$$f''(1) = 6(1 - 2) = -6 < 0 \implies \text{Concave Down} \implies \boxed{\text{Local Maximum}}$$

$$y_{\max} = f(1) = 1^3 - 6(1)^2 + 9(1) = 4$$

2. At $x_2 = 3$:

$$f''(3) = 6(3 - 2) = +6 > 0 \implies \text{Concave Up} \implies \boxed{\text{Local Minimum}}$$

$$y_{\min} = f(3) = 3^3 - 6(3)^2 + 9(3) = 27 - 54 + 27 = 0$$

Task 3: Inflection Point

Setting $f''(x) = 0$:

$$6(x - 2) = 0 \implies x_{\inf} = 2$$

Evaluating $f(2)$:

$$f(2) = 2^3 - 6(2)^2 + 9(2) = 8 - 24 + 18 = 2$$

The inflection point is located at $(2, 2)$, where the slope is:

$$f'(2) = 3(2)^2 - 12(2) + 9 = 12 - 24 + 9 = -3$$

Task 4: Second-Order Taylor Polynomial at $x = 3$

The second-order Taylor expansion around $x_0 = 3$ is:

$$P_2(x) = f(3) + f'(3)(x - 3) + \frac{f''(3)}{2!}(x - 3)^2$$

Substitute known values $f(3) = 0$, $f'(3) = 0$, $f''(3) = 6$:

$$P_2(x) = 0 + 0(x - 3) + \frac{6}{2}(x - 3)^2 = \boxed{3(x - 3)^2}$$

This parabola matches the curvature and local minimum of $f(x)$ at $x = 3$, demonstrating how Newton's optimization method uses quadratic approximations to step directly to optimal points.